

CSCI 112

Programming with C

LECTURE: 1

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OUTLINE:

??

SYLLABUS

INTEGRATED DEVELOPMENT ENVIRONMENT

An Integrated Development Environment (IDE) is a comprehensive software application that consolidates the essential tools needed for software development into a single graphical user interface. It typically combines:

1. a source **code editor** for writing code
2. a **compiler** and **linker** to transform that code into an executable program.
3. a built-in **debugger** to help developers identify and resolve logical errors

efficiently during the programming process.

INTEGRATED DEVELOPMENT ENVIRONMENT



```
int *ptr = malloc(sizeof(int));  
if (ptr != NULL) {  
    *ptr = 112;  
    free(ptr);  
    free(ptr);  
}
```



EXAMPLE

description

description

description

description

Output

Hello world

DLACZEGO C W 2026?

Wydajność (Theory)

Minimalna abstrakcja.

Bezpośrednie mapowanie na instrukcje procesora.

Standardy Montana Tech.

PRZYKŁADOWY KOD

```
int *ptr = malloc(sizeof(int));  
if (ptr != NULL) {  
    *ptr = 112;  
    free(ptr);  
}
```



TITTLE

High Performance

C provides minimal abstraction.

Code maps almost directly to machine instructions,

ensuring the fastest execution for heavy computational tasks.

C provides minimal abstraction.

```
int *ptr = malloc(sizeof(int));  
if (ptr != NULL) {  
    *ptr = 112;  
    free(ptr);  
}
```



ARRAY

Fruit	Colour	Amount	Cost
Banana	Yellow	4	£1.00
Apple	Red	2	£0.60
Orange	Orange	10	£2.50
Coconut	Brown	1	£1.50

THANK

You